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CMSC 204 UserAccessManager Project 1 Pseudocode

// pseudocode for UserAccount.java

Declare a public class UserAccount

Declare a private String variable username

Declare a private String variable encryptedPassword

Declare a private integer variable failureCount initialized to 0

Declare a private boolean variable locked initialized to false

Declare a public constructor UserAccount with String parameter arguments username, encryptedPassword

Set this.username equal to username

Set this.encryptedPassword equal to encryptedPassword

Set failureCount to 0

Set locked to false

// end of constructor

Declare a public void method incrementFailureCount with no parameter arguments

 If locked is false

 Increment failureCount by 1

 // end of if statement

 If failureCount is greater than 2

 Set locked to true

 // end of if statement

// end of incrementFailureCount method

Declare a public void method resetFailureCount with no parameter arguments

 Set failureCount to 0

// end of resetFailureCount method

Declare a public boolean method isLocked with no parameter arguments

```
    Return the value of locked  
// end of isLocked method
```

Declare a public boolean method checkPassword with a String parameter argument password

```
    If locked is true  
        Throw AccountLockedException with message "User  
'username' account is locked."  
    // end of if statement  
    If encryptedPassword does not equal password  
        Throw PasswordIncorrectException with message "Incorrect  
Password"  
    // end of if statement  
    Return true  
// end of checkPassword method
```

Declare a public boolean method equals with Object parameter argument object

```
    If this equals object  
        Return true  
    // end of if statement  
    If object is null or class of object is not equal to class of this  
        Return false  
    // end of if statement  
    Cast object to UserAccount named otherUser  
    If this.username equals otherUser.username  
        Return true  
    Else  
        Return false  
    // end of if-else statement  
// end of equals method
```

Declare a public String method toString with no parameters

Return username

// end of toString method

Declare a public int method getFailureCount with no parameters

Return failureCount

// end of getFailureCount method

// end of class UserAccount and UserAccount.java pseudocode

// UserAccessManager.java pseudocode

Declare a public class UserAccessManager

Declare a private List object of UserAccount named accounts

Declare a public constructor UserAccessManager with no parameter arguments

Initialize accounts as a new ArrayList of UserAccount

// end of constructor

Declare a public void method loadAccounts with String parameter argument filename

Call Utilities.readAccountFile with parameter arguments filename and this

// end of loadAccounts method

Declare a public void method addUser with String parameter arguments username, encryptedPassword

If username is null or username is empty or encryptedPassword is null or encryptedPassword is empty

Throw InvalidCommandException with message "Invalid username or password."

// end of if statement

Declare a UserAccount variable newUser and initialize with username and encryptedPassword

```
    If accounts contains newUser
        Throw DuplicateUserException with message "User 'username'
account already exists."
    // end of if statement
    Add newUser to accounts
// end of addUser method
```

Declare a public void method removeUser with String parameter argument username

```
    If username is null or username is empty
        Throw InvalidCommandException with message "Invalid
username."
    // end of if statement
```

Declare a UserAccount variable temporaryAccount and initialize with username and any placeholder password

```
    If accounts.remove(temporaryAccount) is false
        Throw UserNotFoundException with message "User 'username'
not found."
    // end of if statement
// end of removeUser method
```

Declare a public boolean method verifyAccess with String parameter arguments username, encryptedPassword

```
    If username is null or username is empty
        Throw InvalidCommandException with message "Invalid
username."
    // end of if statement
    If encryptedPassword is null or encryptedPassword is empty
        Throw InvalidCommandException with message "Invalid
password."
    // end of if statement
```

```
For each UserAccount user in accounts
    If user.toString() equals username
        Try
            If user.checkPassword(encryptedPassword) is true
                Call user.resetFailureCount
                Return true
            // end of if statement
        Catch a PasswordIncorrectException parameter exception
            Call user.incrementFailureCount
            If user.isLocked() is true
                Throw AccountLockedException with message "User
'username' account is locked."
            // end of if statement
            Throw exception
        // end of try-catch block
    // end of if statement
// end of for loop
Throw UserNotFoundException with message "User 'username'
not found."
// end of verifyAccess method
```

```
// end of class UserAccessManager and UserAccessManager.java
pseudocode
```

```
// pseudocode for AccountLockedException.java
```

```
Declare a public class AccountLockedException that extends Exception
```

```
Declare a public constructor AccountLockedException with a String
parameter argument message
```

```
    Call superclass with parameter message
// end of constructor
```

```
// end of class AccountLockedException and
AccountLockedException.java pseudocode
```

// pseudocode for DuplicateUserException.java

Declare a public class DuplicateUserException that extends Exception

Declare a public constructor DuplicateUserException with a String parameter argument message

 Call superclass with parameter message

// end of constructor

// end of class DuplicateUserException and
DuplicateUserException.java pseudocode

// pseudocode for InvalidCommandException.java

Declare a public class InvalidCommandException that extends
Exception

Declare a public constructor InvalidCommandException with a String parameter argument message

 Call superclass with parameter message

// end of constructor

// end of class InvalidCommandException and
InvalidCommandException.java pseudocode

// pseudocode for PasswordIncorrectException.java

Declare a public class PasswordIncorrectException that extends
Exception

Declare a public constructor PasswordIncorrectException with a String parameter argument message

 Call superclass with parameter message

// end of constructor

// end of class PasswordIncorrectException and
PasswordIncorrectException.java pseudocode

// pseudocode for UserNotFoundException.java

Declare a public class UserNotFoundException that extends Exception

Declare a public constructor UserNotFoundException with a String
parameter argument message

 Call superclass with parameter message

// end of constructor

// end of class UserNotFoundException and
UserNotFoundException.java pseudocode